Kuta Software - Infinite Calculus

Name___________________________________

Derivatives of Inverse Functions

Date_________________ Period____

For each problem, find $\left(f^{-1}\right)'(x)$ by direct computation.

1) $f(x) = -3x + 3$

2) $f(x) = -2x + 3$

For each problem, find $\left(f^{-1}\right)'(x)$ by using the theorem $\left(f^{-1}\right)'(x) = \dfrac{1}{f'\left(f^{-1}(x)\right)}$

3) $f(x) = -5x + 1$

4) $f(x) = -2x + 2$

5) $f(x) = \sqrt{-2x - 3}$

6) $f(x) = -4x^3 - 4$

For each problem, find $\left(f^{-1}\right)'(x)$ by using the formula $\dfrac{dy}{dx} = \dfrac{1}{\frac{dx}{dy}}$, where $y = f^{-1}(x)$

7) $f(x) = x^7 + x - 3$

8) $f(x) = 3x^5 + 2x + 5$